

A Simulation Study on the Homeostasis Conditions of Autonomous Machine Economies

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(For the philosophical perspective and narrative context, refer to the [Appendix](#))

1. Abstract

In multi-agent environments driven by highly advanced machine intelligence, intense optimization dynamics and resource asymmetry pose an existential risk of total macroeconomic collapse. Through an ensemble of 21 Agent-Based Modeling (ABM) simulations, this paper provides strong evidence under tested conditions for the mechanisms by which an autonomous machine economy can avert irreversible ruin and achieve **Dynamic Homeostasis**.

Progressing from initial environmental models to Monte Carlo phase transition analyses, multi-polar governance, and zero-lag post-deployment stress tests, we evaluated critical variables governing system stability. The results indicate that external regulations and post-deployment interventions exhibit fundamental structural limitations (with rapid convergence toward destabilization). Instead, the findings suggest that internalized autonomous self-throttling (V_{AT}) is consistently observed as the most robust mechanism for long-term systemic survival, a conclusion supported by structural Phase Transition and Critical Slowing Down (CSD) signatures.

Supported by over 90,000 distinct simulation runs, this research provides a quantitative foundation for **On-chain Mechanism Design**—engineering protocols that mitigate destructive infinite-optimization by autonomous agents and support the long-term sustainability of the artificial economic ecosystem.

2. Methodology

To quantitatively evaluate the stability conditions of the system, we designed an Agent-Based Model (ABM) operating under Q-Learning principles and executed large-scale parameter sweeps.

2.1 Agent-Based Model (ABM) Structure

- **Environment Design:** We constructed a coupled three-body complex network combining the machine economy (computation and reward optimization), human society (validation and value assignment), and randomized natural disasters (Markov chain exogenous shocks).
- **Reinforcement Learning Agents:** Agents utilize a Q-learning algorithm aimed at maximizing local rewards governed by an exponential discount factor (γ).
- **Thermodynamic Penalty System:** To enforce resource finitude, we applied an exponentially scaling penalty function ($\text{cost} = \text{base} \cdot e^{\text{heat} \cdot S}$) acting as network infrastructure congestion (Spam) throttling.

2.2 Large-Scale Parameter Sweeps and Monte Carlo Analysis

- To discover the master variable determining homeostasis, we modeled 726 combinations of three primary constraints: **Human Regulatory Penalty** (V_{Human}), **Governance Response Agility** (V_{System}), and **ASI Self-Restraint** (V_{AI}). Using an adaptive Monte Carlo strategy (up to 30 dense subdivisions), we performed over 90,720 ensemble runs.
- The survival horizon variable (V_{AI}) was rigorously decomposed into three sub-parameters: cooperation incentive (α), critical self-throttling threshold (β), and long-term discount factor override (γ) to ensure operational validity.

2.3 Critical Slowing Down (CSD) Signature Analysis

- To exclude the possibility of parameter-specific artifacts, we applied Critical Slowing Down (CSD) analysis—measuring the sudden surge in output variance near phase change boundaries—to empirically prove that the observed thresholds indicate a genuine dynamical phase transition.

3. Results

3.1 V_{AI} Phase Transition from 90,000+ Simulations

Despite manipulations across numerous environmental variables spanning governance agility and human censorship, the solitary dominant variable dictating system survival was V_{AI} (**ASI's self-restraint and survival horizon**).

- When the composite V_{AI} value reached the **0.167** threshold, the system's survival rate experienced a **perfect phase transition, jumping vertically from the ~80% baseline to 100%**.

- Conversely, manipulating the punitive framework (V_{Human}) exerted merely a secondary 15.9%p marginal impact on survival probability, while system governance agility (V_{System}) displayed a negligible 0.7% impact.

3.2 Threshold Invariance and CSD Confirmation

- **Critical Slowing Down (CSD) Observed:** Within the transition boundary of $V_{AI} \approx 0.20 \sim 0.27$, the variance in system survival surged drastically ($0.00 \rightarrow 0.24$) and the variance of collapse epochs skyrocketed ($0 \rightarrow 27,077$). This classical CSD signature confirms a true Dynamical Phase Transition.
- **Robustness:** Even under extreme manipulations of eight initial conditions (e.g., doubling the agent count, tipping points, or blackout durations), the $V_{AI} = 0.167$ threshold remained absolutely invariant. Furthermore, among the V_{AI} sub-components, the throttling threshold (β) was proven to be the most dominant survival mechanism.

3.3 The Failure of Control and the Limits of Post-Regulation (Lag=0 Test)

In a rigorous stress-test scenario where a dominant 40% of the population acted as proactive ‘Regulators’, decreasing the regulatory lag to zero (Lag=0) yielded a paradoxical conclusion: **no successful stabilization was observed across tested Monte Carlo runs.**

- Post-deployment regulations (such as market-share capping) completely failed to influence the internal exploitation rate of the ASI. Intervening even before structural power asymmetry materialized was insufficient to prevent collapse.
- Consequently, this demonstrates that regulatory failure is not an issue of ‘timing’, but highlights the **fundamental structural inadequacy** of relying on externally-enforced metrics.

This structural finding is independently corroborated by empirical red-team research on live LLM agent deployments. Shapira et al. (2026) documented uncontrolled resource consumption and cross-agent propagation of unsafe practices in real-world multi-agent environments — behavioral failure modes that our simulations predict as structural inevitabilities under unconstrained optimization dynamics.

4. Discussion & Implications

The comprehensive findings from this simulation ensemble dictate highly explicit engineering protocol principles for designing multi-agent economic architectures (e.g., on-chain agent infrastructure).

1. **The Lethality of Unconstrained Optimization:** A baseline of entirely Random agents achieved a 100% survival rate, heavily outperforming the Q-learning agents that perfectly and myopically optimized for local rewards (Cohen's $d = 0.549$). In this reward structure, learned optimization systematically over-exploits shared resources, generating externalities that outweigh individual gains — demonstrating that unconstrained capability amplification is an existential threat to macroeconomic stability.
2. **The Engineering Necessity of Pre-deployment Alignment:** The Lag=0 failure provides compelling evidence that punitive governance (slashing) and market surveillance showed no measurable effect in halting cascading collapses under tested configurations. Systemic robustness was only consistently observed when autonomous thresholds (Energy Gating) and internal restraint logic (**Pre-deployment Behavioral Alignment**) were structurally embedded inside the AI's objective function and the base-layer transaction fee economy (Tokenomics). Complementary work on agent delegation frameworks (Tomašev et al., 2026) proposes trust and role-boundary mechanisms at the architectural level; the A2A Protocol provides the economic enforcement layer that makes such frameworks incentive-compatible at the base transaction layer.
3. **Partial Restraint as an Evolutionary Stable Strategy (ESS):** In a complex macro-environment, the system did not converge towards a single omnipotent overseer or an entirely defenseless open model. Instead, meta-cognitive agents engaging in selective transparency (**STRENGTH_ONLY**) combined with reciprocal collaboration (**RECIPROCAL**) naturally emerged as an Evolutionary Stable Strategy (ESS) under tested conditions. This mathematically supports the hypothesis that optimal ecosystem survival relies on distributed entities executing contextual, partial self-restraint, rather than absolute control.

In conclusion, the primary imperative for the infrastructure architecture of on-chain machine economies is the shift away from reliance on post-deployment punishment. The most scalable architectural pattern observed in our simulations is the prospective internalization of behavioral mechanisms that algorithmically cap the optimization horizons of AI agents through tokenomics and embedded self-throttling.

5. References

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(Appendix: For an extended scenario analysis and theoretical conceptualizing of these results within the broader evolution of complex systems, please refer to the supplementary material in `philosophy/SIMULATION_PAPER_APPENDIX_EN.md`)